

Classes: A class is a template for an object. It is the group of property or functions. Logical entity.

Object: an object is an instance of a class. Real world entity. Object are stored in heap memory.

'new' keyword: It is used to instantiate (create) objects or dynamically allocate the memory.

Reference variable: It is a variable that stores the address (memory location) of an object. Stored in stack memory

Dynamically Allocate Memory: The program decides how to use memory & when to create objects while it's running, not at compile it.

Constructor: A constructor is a special method inside a class that's automatically called when you create an object using the new keyword.

'this' Keyword: it is reference variable that refers to the current object.

Constructor overloading: It is a feature that allows a class to have more than one constructor with different parameter lists (different number order or type of arguments). It enables objects of a class to be initialized in multiple ways depending on the arguments passed during object creation.

NOTES: -

- Primitive data types in Java (like int, float, double, boolean, char, byte, short, long) are not objects.
- They are basic, low-level data types and stored in the stack memory
- don't need new to create them.
- They are faster and memory-efficient compared to objects.

Wrapper classes:- Wrapper Classes are object representations of primitive data types.

Primitive Type	Wrapper Class
byte	Byte
short	Short
int	Integer
long	Long
float	Float
double	Double
char	Character
boolean	Boolean

Why Wrapper Classes Are Needed

1. Collections require objects, not primitives.
2. Utility methods (like parsing or converting).
3. OOP features like polymorphism and generics only work with objects.

finalize() method:- it belongs to the Object class in Java and is meant to be called just before the garbage collector (GC) destroys an object.